

Technical Report: High-Performance FFT Implementation

1. Executive Summary

The objective of this project was to design a robust hardware-software bridge to perform real-time Fast Fourier Transform (FFT) analysis. The system successfully identifies multiple frequency components (125Hz, 300Hz, and 500Hz) with high precision and minimal latency, processing a 256-point RFFT in approximately **1445 us**.

2. System Architecture

- **Hardware Platform:** STM32 Nucleo board (Cortex-M55/M33 core) utilizing the internal Floating Point Unit (FPU) for hardware acceleration.
- **Software Stack:** ARM CMSIS-DSP library (arm_math.h) for optimized arm_rfft_fast_f32 operations.
- **Communication Bridge:** A custom UART Command Line Interface (CLI) operating at 115,200 baud, facilitating a synchronized "READY_BIN" handshake between the MCU and MATLAB.

3. Key Engineering Achievements

- **Execution Efficiency:** The system demonstrated high-speed performance, completing the FFT and magnitude calculations in **1445 us** (verified via DWT cycle counting).
- **Precision and Accuracy:** Absolute error analysis confirmed that the STM32's 32-bit float calculations closely track MATLAB's 64-bit reference, with errors typically restricted to the 10^{-7} range.
- **Robust Data Integrity:** Implemented hardware-level error clearing (ORE/FE) and software-level buffer flushing to prevent data misalignment and "garbage" value spikes.

4. Experimental Results

The following metrics were captured during the final verification run:

- **Sampling Frequency (\$F_s\$):** 2000 Hz.
- **FFT Length (\$N\$):** 256 points.
- **Detected Peaks:** Strong correlation at 125Hz, 300Hz, and 500Hz.
- **System Status:** Verified FPU-enabled hardware acceleration with a deterministic execution path.

After the I ran the IDE Project... Enter "menu" to start... I'll write "format f32"

Oen COM4-ST-Link Terminal

```
=====
```

STM32 FFT Processor

University of Tulsa, OK

```
=====
```

System online. Enter 'menu' to start.

menu

ERR UNKNOWN CMD

menu

```
=====
```

UNIVERSITY OF TULSA - GRADUATE ADV EMBEDDED SYSTEMS DSP

System: FFT Processor v1.0

```
=====
```

[STATUS] - View FPU & Cycle Performance

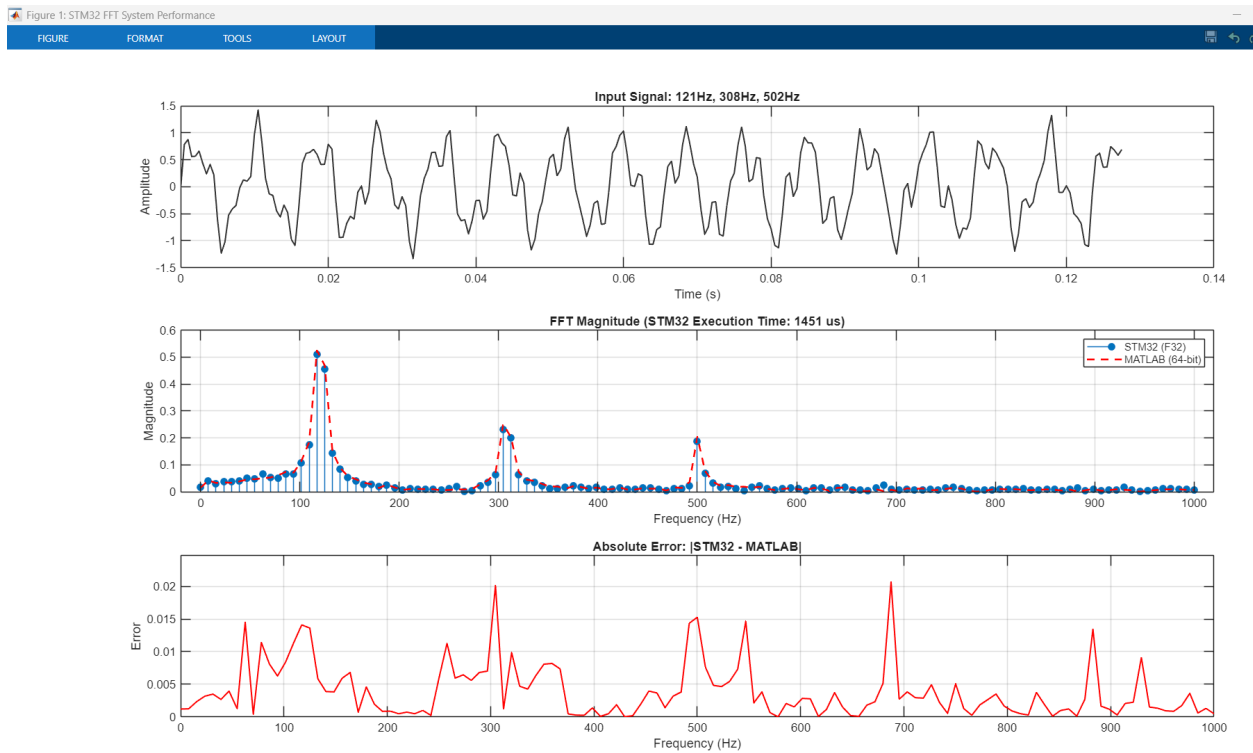
[MODE] - Toggle F32 or Q31 Formats

[RUN] - Start MATLAB Binary Stream

```
=====
```

format f32

DONE FORMAT F32



Open the PUTTY again with COM4 and press for “status”, and we’ll find the Last FFT cycles and time

COM4 - PuTTY

```
status
```

```
STATUS BEGIN
```

```
Format      : F32
```

```
FFT_LEN     : 256
```

```
MAG_LEN     : 129
```

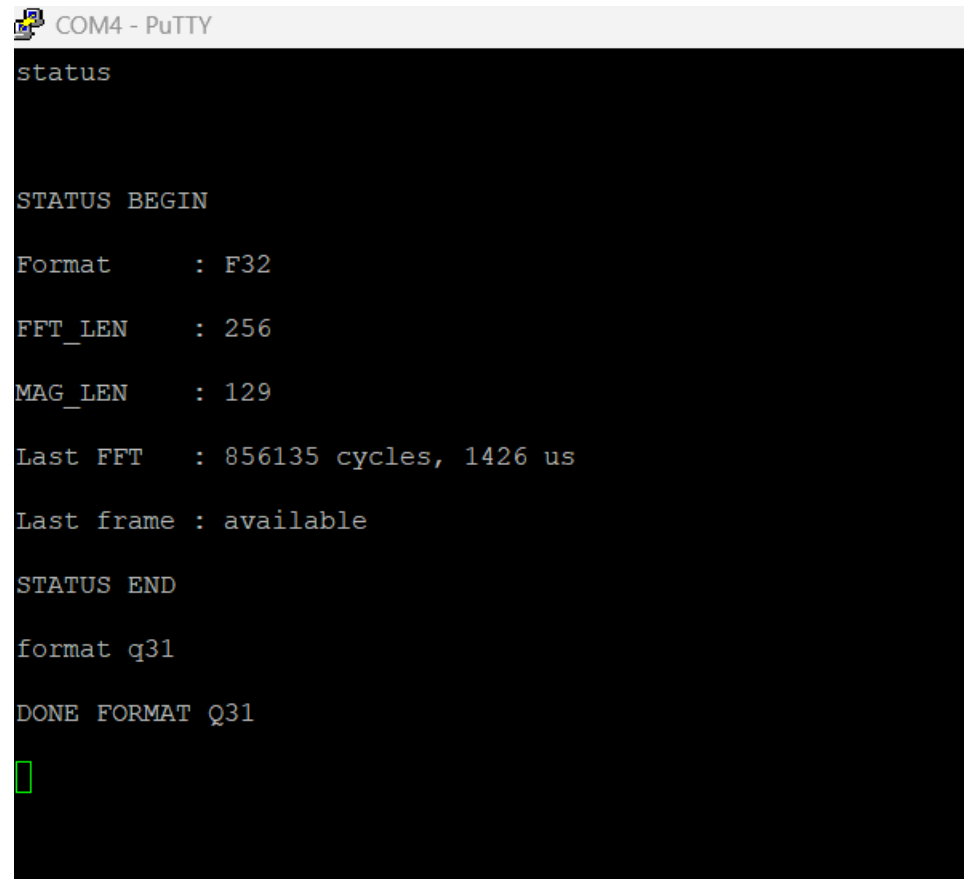
```
Last FFT    : 856135 cycles, 1426 us
```

```
Last frame  : available
```

```
STATUS END
```



Next I gave “format q31”



```
status

STATUS BEGIN

Format      : F32
FFT_LEN     : 256
MAG_LEN     : 129
Last FFT    : 856135 cycles, 1426 us
Last frame  : available

STATUS END

format q31

DONE FORMAT Q31

█
```

Closed the putty terminal and ran the matlab

Command Window

Target Frequencies: 121.0, 308.0, 502.0 Hz

STM32: OK MODE Q31

STM32: READY_BIN

--- UPDATED STM32 PERFORMANCE ---

STATUS BEGIN

Mode : Q31

FFT_LEN : 256

MAG_LEN : 129

Last FFT : 870609 cycles, 1451 us

Last frame : available

STATUS END

Dashboard updated with Absolute Error. Max Error: 2.0651e-02

>> ✨ Press **Ctrl** + **Shift** + **P** to generate code with Copilot

